

# Hyunah Lim

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## EDUCATION

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**Ph.D. in Mathematics**, University of Maryland, 2019-2026.

Dissertation defended December 2025; degree expected Spring 2026.

**M.S. in Mathematics**, Yonsei University, 2017-2019.

**B.A. Double Major in Economics and Applied Statistics, Minor in Mathemtaics**,  
Yonsei University, 2012-2017.

## PREPRINTS AND PUBLICATIONS

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**Evolution of drug resistance in age-structured tumor growth model.**

Doron Levy and Hyunah Lim

In preparation

**Age-structured mechanical models for tumor growth.**

Doron Levy, Hyunah Lim, Antoine Mellet, and Maeve Wildes

Submitted, [Full manuscript](#)

**Mathematical assessment of the roles of vaccination and Pap screening on the burden of HPV and related cancers.**

Soyoung Park, Hyunah Lim, and Abba B. Gumel

*Bulletin of Mathematical Biology* **87**(12) (2025), 182. [DOI:10.1007/s11538-025-01548-5](https://doi.org/10.1007/s11538-025-01548-5)

## THESES

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**Age-structured models of tumor growth and evolution of drug resistance.**

Ph.D. dissertation University of Maryland, Department of Mathematics, 2025

**Some basic understandings of the Lascar Group: Lascar Group is a topological group.**

Master's thesis Yonsei University, Department of Mathematics, 2019 [RISS](#)

## TALKS

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*Age-structured models of tumor growth and evolution of drug resistance.* Doctoral Dissertation Defense, University of Maryland, Dec 2025.

*Age-structured mechanical models for tumor growth.* RIT on Mathematics of Infectious Diseases, University of Maryland, Nov 2025.

*Mathematical modeling of the impacts of screening and vaccination in HPV* (Poster talk).

Mathematical Opportunities in Digital Twins, George Mason University, Dec 2023.

*Mathematical modelling of the impacts of screening and vaccination in HPV* (Poster talk). Predictive Modeling in Biology and Medicine Conference, University of California, Riverside, Nov 2023.

*Introducing approaches to tumour growth models*. Preliminary Oral Examination, University of Maryland, Mar 2023.

*Lascar group is a topological group*. Master's Thesis Defense Presentation, Yonsei University, Dec 2018.

## AWARDS

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- Ruth-Davis Fellowship, University of Maryland, 2019-2024
- Dean's Fellowship, University of Maryland, 2019-2021.
- Academic Excellence Scholarship, Yonsei University, 2015

## TEACHING EXPERIENCE

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### University of Maryland

- Instructor, Math 120 - Elementary Calculus I (Current, Summer 2025)
- Teaching Assistant, Math 240 - Linear Algebra (Fall 2021, Fall 2023, Fall 2024, Spring 2025, Fall 2025).
- Teaching Assistant, Math 120 - Elementary Calculus I (Fall 2019, Spring 2020, Spring 2022, Spring 2024).
- Teaching Assistant, Math 140 - Calculus I (Spring 2023)
- Teaching Assistant, Math 141 - Calculus II (Fall 2020, Fall 2022).
- Teaching Assistant, Math 115 - Precalculus (Spring 2021).

### Yonsei University

- Grader, MAT 2104 - Set Theory (Spring 2019)
- Teaching Assistant, MAT3109 - Abstract Algebra I (Spring 2018)
- Teaching Assistant, MAT1012 - Mathematics for Engineers II (Spring 2018, Fall 2018)
- Teaching Assistant, MAT1002 - Calculus and Vector Analysis II (Fall 2017)
- Teaching Assistant, MAT1001 - Calculus and Vector Analysis I (Spring 2017, Fall 2018)
- Teaching Assistant, MAT1011 - Mathematics for Engineers I (Spring 2017, Fall 2017)

## LANGUAGES

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English (fluent); Korean (fluent); Japanese (Basic).